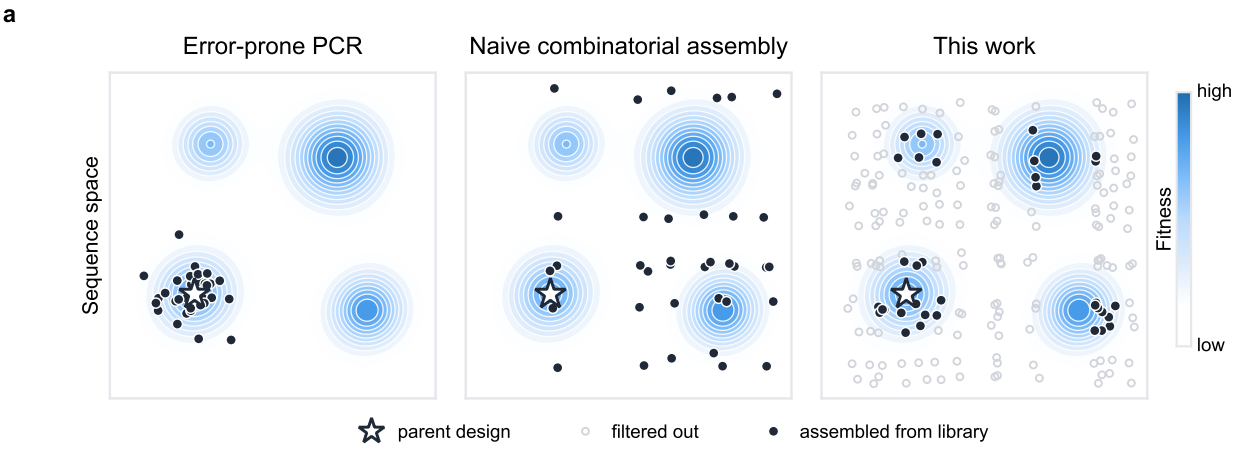
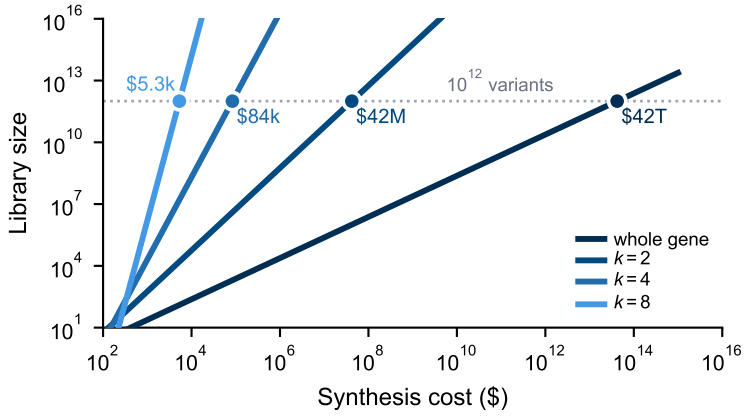
Design of high fidelity combinatorial libraries

November 5, 2026

Abstract

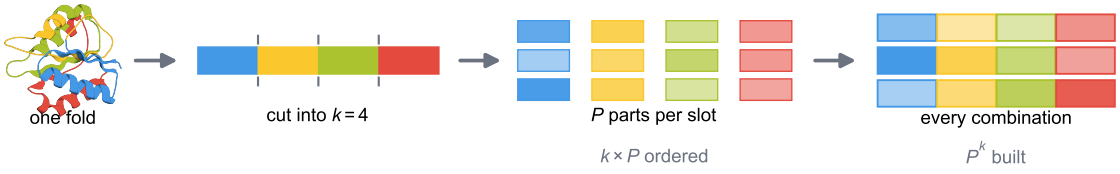
Recent tools for de novo design have revolutionized the way we propose novel proteins and the types of challenges we are able to tackle. While these tools have dramatically increased the hit rate of functional designs, zero-shot generations often require further optimization to enhance the desired properties. We present a novel method for generating high fidelity diverse combinatorial libraries. Traditional library generation methods rely on error prone PCR or natural fragment shuffling to generate diversity. Such approaches do not take into account the filtering of designs through structure prediction models or other desired oracles. We demonstrate the ability to post-train inverse folding models to design fragments with filter pass rates of 93.4% and libraries which can easily scale beyond 10^{12} individually addressable variants. Our post-training framework enables control over the search space by centering the library around a reference sequence with tunable mutation distance. Once designed, we demonstrate how individual variants are assembled with simple molecular biology techniques (golden gate assembly). Our cell-free assay platform enables us to screen enzymes at the scale of 10^3 measurements per day, each with replicates. The libraries generated are easily much larger than most traditional and modern high throughput assays are able to screen, so we demonstrate how the library can be sub sampled smartly to optimize enzymes. For a de novo designed PETase we demonstrate our platform finds a variant with 250x increase in catalytic activity.





c

panel (c) reserved



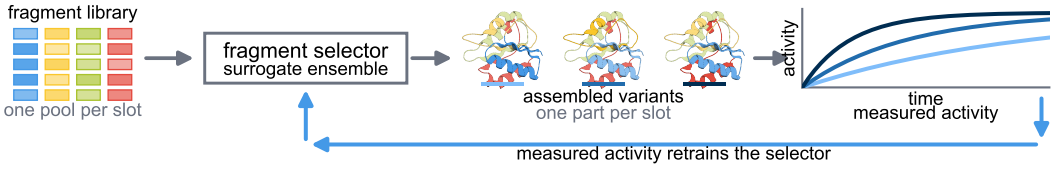
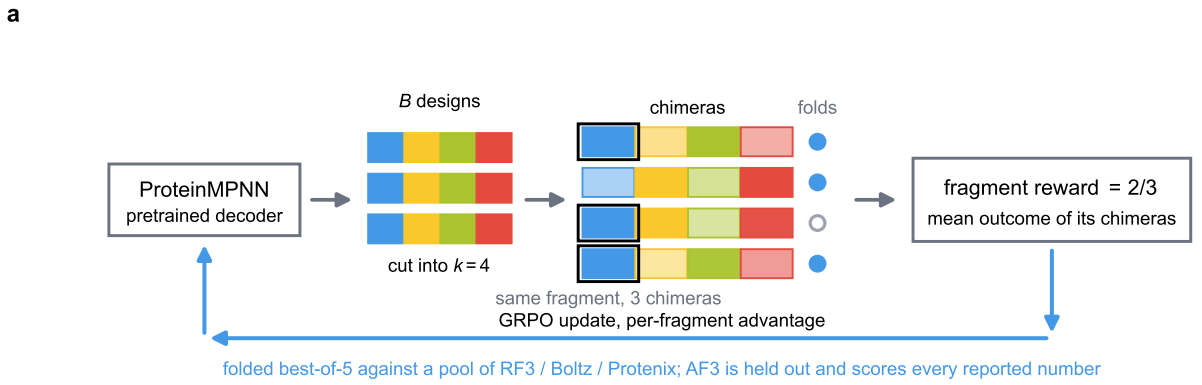
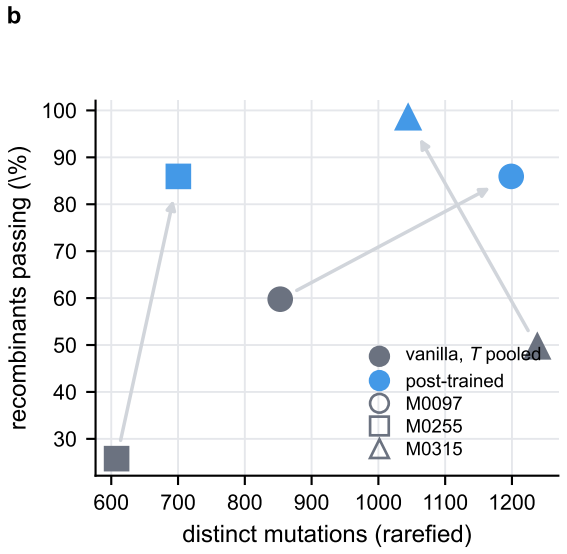
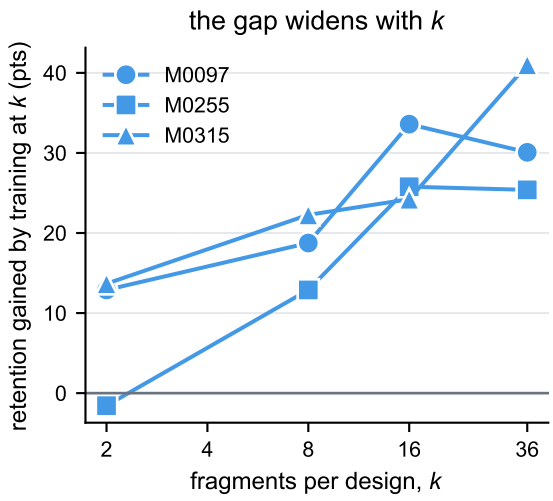


Figure 1: **A combinatorial library, and the loop around it.** [PLACEHOLDER — the prose is yours. Kept short so the float fits; the facts below are the ones a reader cannot get from the picture.] **(a)** One toy landscape searched three ways with the same number of variants built and screened ($N = 36$). Illustrative: no axis is measured. **(b)** eBlocks are billed at a 300-bp minimum, which binds from $k = 2$ on, so past that point cutting finer stops buying shorter DNA and only buys more pieces of it. 10^{12} variants costs \$42T as whole genes, \$42M at $k = 2$, \$84k at $k = 4$, \$5.3k at $k = 8$. **(c)** [Reserved — to be filled in.] **(d)** What makes the library combinatorial. One fold is cut at $k - 1$ boundaries; each slot is then an independent pool of P parts; and any one part per slot assembles a full-length design. Ordering $k \times P$ fragments therefore builds P^k sequences, which is the exponent panel (b) prices. Parts per slot are drawn as three to keep the columns legible. **(e)** The experimental loop. The arrow returns to the *selector*, not the library: parts are synthesised once and each round changes only which of them get combined.


a
B designs
chimeras
folds
ProteinMPNN
pretrained decoder
cut into k = 4
fragment reward = 2/3
mean outcome of its chimeras
same fragment, 3 chimeras
GRPO update, per-fragment advantage
folded best-of-5 against a pool of RF3 / Boltz / Protenix; AF3 is held out and scores every reported number


b
recombinants passing (%)
distinct mutations (rarefied)
vanilla, T pooled
post-trained
M0097
M0255
M0315


c
the gap widens with k
retention gained by training at k (pts)
fragments per design, k
M0097
M0255
M0315

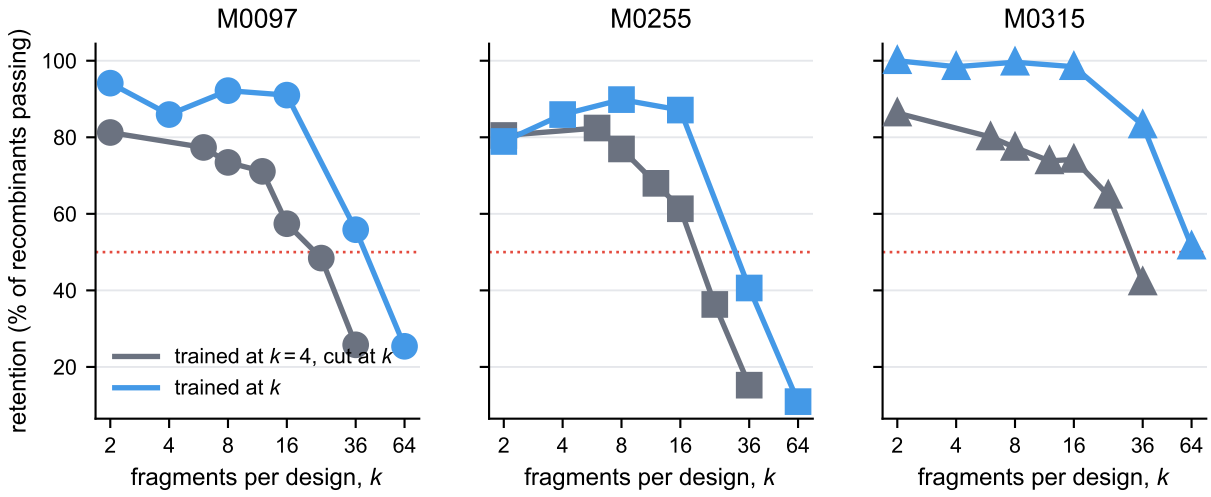

d
M0097
M0255
M0315
retention (% of recombinants passing)
fragments per design, k
trained at k = 4, cut at k
trained at k

Figure 2: **Designing high-fidelity combinatorial libraries.** [PLACEHOLDER — the prose is yours.] **(a)** How post-training works here. ProteinMPNN is pretrained and emits whole sequences, so nothing in it says which fragments *recombine*; the method manufactures that signal. A batch is cut into k slots, chimeras are built by swapping slots between parents, each chimera is folded, and every fragment is given the mean outcome of the chimeras it appeared in — here one fragment occurs in three, two of which fold, so its reward is 2/3. That per-fragment reward, not the parent's score, drives the GRPO update. Folding during training is best-of-5 against a *pool* of RF3, Boltz and Protenix under a round robin; AlphaFold3 is held out of training entirely and scores every number in (b)–(d). **(b)** Stock ProteinMPNN against the post-trained policy, for the three backbones trained against the pool. The vanilla arm is not a single temperature: its fragments come from parents sampled at $T = 0.1$–1.0 and then filtered to those that pass, which skews the pool toward low T (for example 29%, 23%, 22%, 16%, 9%, 1% of fragment slots on M0097). It is therefore the filtered union of the whole temperature sweep rather than any one setting. Both arms are measured on the same thing: sequences decoded from MPNN, cut at $k = 4$, shuffled across parents, reassembled, and folded — never on parents. Arrows run from vanilla to post-trained. Success rises on all three, by +26.2, +60.2 and +48.8 points. Distinct mutations are counted over *passing* recombinants only and rarefied to a matched survivor count, so a wider spread is genuinely wider coverage and not just a larger sample; on that measure two backbones move right as well as up and M0315 trades 193 mutations for its 48.8 points. **(c)** How much of the retention in (d) is owed to training at that k rather than to cutting a $k = 4$ policy finer: +8.3 points at $k = 2$, rising monotonically to +32.2 at $k = 36$. **(d)** Retention against the number of fragments per design. Blue: a policy trained at that k. Grey: one $k = 4$ policy re-cut at k post hoc. Every trained cell ran a matched 200 steps so budget cannot covary with k. Caveats that must survive into the text: every cell is single-seeded against a documented 21-point run-to-run swing; M0255 at $k = 36$ and $k = 64$ was still climbing at step 200, so those are lower bounds; and the vanilla arm in (b) is one pooled recombinant set per backbone, not a per-temperature curve.